HDOC Day-42

String Operations Using a Menu-Driven C Program

Aim

To write a menu-driven C program using separate user-defined functions to remove vowels, find the first repeating lowercase alphabet, check whether two strings are anagrams, and exit.

Algorithm

1. Start the program and display the four-option menu.

2. Accept the user choice and use a switch statement to call the corresponding function.

3. To remove vowels, read a string, traverse it, and copy only non-vowel characters into another string.

4. To find the first repeating lowercase alphabet, count frequencies of lowercase letters, then scan from left to right for the first letter occurring more than once.

5. To check anagrams, read two strings, convert letters to lowercase, count each letter in both, and compare their frequency arrays.

6. For Exit, terminate the program. Show an error for invalid choices and repeat until Exit is selected.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
#define MAX 200
void removeVowels(void);
void firstRepeatingLowercase(void);
void checkAnagram(void);
int main()
{
    int choice;
    do
    {
        printf("\n----- STRING OPERATIONS MENU -----\n");
        printf("1. Remove All Vowels\n");
        printf("2. Find First Repeating Lowercase Alphabet\n");
        printf("3. Check Anagram\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();
        switch (choice)
        {
            case 1:
                removeVowels();
                break;
            case 2:
                firstRepeatingLowercase();
                break;
            case 3:
                checkAnagram();
                break;
            case 4:
                printf("Exiting program...\n");
                break;
```

```
            default:
                printf("Invalid choice! Please enter 1 to 4.\n");
        }
    } while (choice != 4);
    return 0;
}
/* Function to remove vowels */
void removeVowels(void)
{
    char str[MAX], result[MAX];
    int i, j = 0;
    char ch;
    printf("Enter a string: ");
    fgets(str, MAX, stdin);
    for (i = 0; str[i] != '\0'; i++)
    {
        ch = tolower((unsigned char)str[i]);
        if (ch != 'a' && ch != 'e' &&
            ch != 'i' && ch != 'o' &&
            ch != 'u')
        {
            result[j] = str[i];
            j++;
        }
    }
    result[j] = '\0';
    printf("String after removing vowels: %s", result);
}
/* Function to find first repeating lowercase alphabet */
void firstRepeatingLowercase(void)
{
    char str[MAX];
    int freq[26] = {0};
    int i;
    printf("Enter a string: ");
    fgets(str, MAX, stdin);
    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            freq[str[i] - 'a']++;
        }
    }
    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            if (freq[str[i] - 'a'] > 1)
            {
                printf("First repeating lowercase alphabet: %c\n",
                       str[i]);
                return;
            }
        }
    }
    printf("No repeating lowercase alphabet found.\n");
}
/* Function to check anagrams */
void checkAnagram(void)
{
    char str1[MAX], str2[MAX];
    int freq1[26] = {0};
    int freq2[26] = {0};
    int i;
    char ch;
    printf("Enter first string: ");
    fgets(str1, MAX, stdin);
```

```
    printf("Enter second string: ");
    fgets(str2, MAX, stdin);
    for (i = 0; str1[i] != '\0'; i++)
    {
        ch = tolower((unsigned char)str1[i]);
        if (ch >= 'a' && ch <= 'z')
            freq1[ch - 'a']++;
    }
    for (i = 0; str2[i] != '\0'; i++)
    {
        ch = tolower((unsigned char)str2[i]);
        if (ch >= 'a' && ch <= 'z')
            freq2[ch - 'a']++;
    }
    for (i = 0; i < 26; i++)
    {
        if (freq1[i] != freq2[i])
        {
            printf("The strings are not anagrams.\n");
            return;
        }
    }
    printf("The strings are anagrams.\n");
}
```

Test Case Table

Test	Operation	Input	Expected Result
1	Remove Vowels	Hello World	Hll Wrld
2	Remove Vowels	Education	dctn
3	Remove Vowels	Programming	Prgrmmng
4	First Repeating	programming	r
5	First Repeating	apple	p
6	First Repeating	abcde	No repeating lowercase alphabet
7	Anagram	listen / silent	Anagrams
8	Anagram	triangle / integral	Anagrams
9	Anagram	hello / world	Not anagrams
10	Invalid Choice	5	Invalid choice message
11	Exit	4	Program exits

Sample Output

```
----- STRING OPERATIONS MENU -----
1. Remove All Vowels
2. Find First Repeating Lowercase Alphabet
3. Check Anagram
4. Exit

Enter your choice: 1
Enter a string: Hello World
String after removing vowels: Hll Wrld

Enter your choice: 2
Enter a string: programming
First repeating lowercase alphabet: r

Enter your choice: 3
Enter first string: listen
Enter second string: silent
The strings are anagrams.

Enter your choice: 4
Exiting program...
```

Result

The menu-driven C program was implemented using separate user-defined functions, character arrays, loops, conditional statements, and a switch-case statement. It removes vowels, identifies the first repeating lowercase alphabet, checks anagrams using frequency arrays, handles invalid choices, and continues until Exit is selected.